AP CALCULUS AB

Unit 1 • Lesson 1-4 • Notes (filled in)

Lesson 1-4 • Continuity at a Point

Notes (filled in)

Learning goals

- State and use the three conditions for continuity at a point.
- Classify discontinuities as removable, jump or infinite.
- Find constants that make a piecewise function continuous, and remove removable discontinuities.

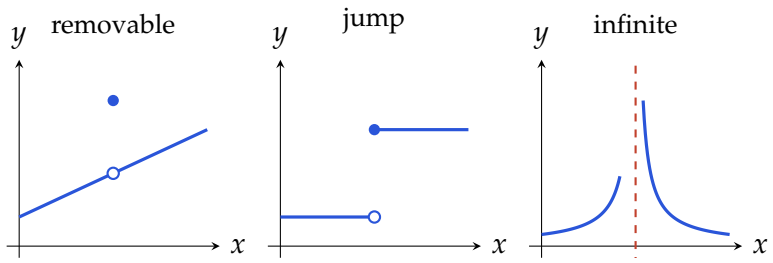
Vocabulary. continuous, removable discontinuity, jump discontinuity, infinite discontinuity, piecewise function.

1. Definition

Key idea

f is continuous at $x = c$ when: (1) $f(c)$ is defined ; (2) $\lim_{x\to c} f(x)$ exists ; (3) $\lim_{x\to c} f(x) = f(c)$.

2. Types of discontinuity



Key idea

Removable: the limit exists but does not equal $f(c)$ (or $f(c)$ is undefined) .
Jump: the one-sided limits exist but are different .
Infinite: at least one one-sided limit is $\pm\infty$.

Example 1 • A complete justification

$f(x) = \begin{cases} x^2 + 1, & x \le 2 \\ 4x - 3, & x > 2. \end{cases}$ Is f continuous at $x = 2$?

$f(2) = 5$. $\lim_{x\to 2^-} f(x) = 5$ and $\lim_{x\to 2^+} f(x) = 5$, so $\lim_{x\to 2} f(x) = 5$. Since $\lim_{x\to 2} f(x) = f(2)$, f is continuous at $x = 2$.

Example 2 • Classify

Classify the discontinuities: $\dfrac{x^2-9}{x-3}$ at 3; $\dfrac{|x|}{x}$ at 0; $\dfrac{1}{x^2}$ at 0.

Removable (limit 6, undefined value). Jump (limits -1 and 1). Infinite ($+\infty$).

3. Making a function continuous

Example 3 • Find the constant

For which k is $f(x)=\begin{cases} kx+1, & x<2 \\ x^2-k, & x\geq 2 \end{cases}$ continuous?

Left limit $2k+1$; right limit and $f(2)$: $4-k$. $2k+1=4-k$, so $k=1$.

Example 4 • Two constants

$g(x)=\begin{cases} ax+b, & x<1 \\ 5, & x=1 \\ x^2+bx, & x>1 \end{cases}$ is continuous at $x=1$. Find a and b.

Right: $1+b=5$, $b=4$. Left: $a+b=5$, $a=1$.

4. Continuity on an interval

Polynomials, rational functions, roots, exponentials, logarithms and trigonometric functions are continuous __on their domains__. Example: $h(x)=\frac{\sqrt{x+2}}{x-1}$ is continuous on __$[-2,1)\cup(1,\infty)$__.

AP exam tip
Model sentence: "$\lim_{x\to 2^-} f(x)=5=\lim_{x\to 2^+} f(x)$, so $\lim_{x\to 2} f(x)=5=f(2)$. Therefore f is continuous at $x=2$."

Check your understanding

1. For $f(x)=\frac{x^2-1}{x-1}$ at $x=1$, condition __(1)__ fails.
2. $\frac{x+2}{x^2-4}$ has a __removable__ discontinuity at $x=-2$ and an __infinite__ one at $x=2$.